

● ---

Summary

1. Expected Tree

```
ai/data/
├─ data.yaml          ← points to cups_labeled/yolo_day1
└─ cups_labeled/yolo_day1/
    ├─ train/
    │   ├─ images/    (≥10 images)
    │   └─ labels/    (1:1 matching .txt, YOLO format, class IDs 0-3)
    ├─ val/
    │   ├─ images/    (≥3 images)
    │   └─ labels/
    └─ test/
        ├─ images/    (≥3 images, SACRED after Step 3)
        ├─ labels/
        ├─ empty/     (10 empty-bin photos, no label files)
        └─ MANIFEST.txt (SHA-256 hashes)
```

2. Actual Tree

```
ai/data/
├─ data.yaml          ← DOES NOT EXIST
└─ cups_labeled/yolo_day1/ ← COMPLETELY EMPTY (0 files)
```

3. Fix: normalize_dataset.py (delivered and tested)

When you have your Roboflow export:

```
source ai/.venv/bin/activate
python ai/src/normalize_dataset.py --source ~/Downloads/your-roboflow-export/
```

It handles:

- valid/ → val/ rename
- Class ID remapping if Roboflow uses a different order (verified with test)
- Copies images + labels into the expected tree
- Writes ai/data/data.yaml
- Creates test/empty/ and test/MANIFEST.txt

- Runs `validate_dataset.py` as the final gate

Both the happy path (same class order) and remap path (different class order) pass end-to-end.